

Question	Marking Guidance	Mark	Comments
08.1	1. A = <u>phospholipid</u> and form a barrier to water-soluble/charged/ions/polar substances OR A = phospholipid and allows lipid soluble/non-polar/hydrophobic substances to pass (through the cell membrane) OR A = <u>phospholipid</u> and simple diffusion; 2. B = <u>protein</u> and allow large/charged/ions/polar molecules to pass (through the cell membrane) OR B = <u>protein</u> and facilitated diffusion OR B = <u>protein</u> and active transport OR B = <u>protein</u> and maintain different concentrations (either side) OR B = <u>protein</u> and providing support/shape;	2 max (2 x AO1)	Both name and function should be correct for 1 mark 1 and 2. Accept hydrophilic for polar Accept for 'water-soluble' non-lipid soluble 2. Ignore the type of protein

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08.2	Mark in pairs: 1 and 2 OR 3 and 4. 1. Volume of water; 2. (Varying volumes of water) affects the concentration/colour (of the resulting solution); 3. Diameter/thickness of disc OR Surface area of disc OR Volume of disc OR Mass of disc OR Same beetroot; 4. Affects quantity/amount of pigment OR Affects rate of release of pigment;	2 max (2 x AO2)	1. Ignore amount of water 3. Allow length for thickness 3. Ignore shape/size 3. Allow weight for mass 3. Accept 'beetroot' for disc

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08.3	MP1 must be credited to achieve maximum 4 marks. 1. As temperature increases, (percentage) concentration of pigment increases but levels off at 80°C; 2. As temperature increases, the membrane becomes damaged/more permeable; 3. Because the membrane proteins denature at high temperatures; 4. As temperature increases, fluidity of the cell membrane/phospholipids increases; 5. At 80°C the membrane is fully permeable OR At 80°C maximum damage to membrane;	4 max (2 x AO1, 2 x AO2)	

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08.4	1. (Use colorimeter to) measure absorbance/transmission of diluted solutions/series/standards; 2. Produce a calibration curve; 3. Measure absorbance/transmission of sample and find concentration (on calibration curve);	3 (3 x AO3)	1. Accept description of absorbance or transmission 2. Accept descriptions of calibration curve